# What survives refinement: the pion, the Yang–Mills gap and Newton's action

**Working conclusion, updated 2026-10-02.** The common problem is to construct a limit while preserving a specified physical structure. Pure $SU(3)$ Yang–Mills asks for a positive energy threshold in a constructed continuum theory. Chiral QCD tests a zero threshold, conditional on broken global symmetry and surviving spectral weight. Newton's comparison asks which independently justified physical premises would force a universal positive action cost of records. The existing record bounds supply quantum kinematics and positive $\hbar$; they establish conditional costs, not the necessity of that input. The quantities, hypotheses and regulators differ in these comparisons.

Two further Newton results sharpen the scope. Fixed-interval Kepler localization and first-moment estimates recover classical trajectories from Coulomb quantum evolution. Relativistic point-Coulomb action thresholds depend on the selected operator domain, including at equality. A coherent-stream record contrast additionally depends on coherence between launch points and the record topology. Neither a chosen domain nor a chosen coherent preparation supplies universal action necessity.

User direction of 2026-09-26 makes this comparison the intended frame of the eventual paper. **The two main proof goals remain the pure $SU(3)$ mass gap and the necessity of a positive action scale in Newton's comparison.** QCD with massless and with nonzero quark masses is an orientation and a source of mechanisms, rather than a third full construction programme. This note supplies the definitions, axial Ward identity and two elementary spectral-limit criteria. It proves neither main goal.

The Planck paper remains the developed Newton component; the Yang–Mills position and conditional theorem remain the technical map for pure gauge theory. Their results retain their stated hypotheses. The October 2 localization, moment and selection additions are written derivations and read source results, without independent verification. This synthesis makes no literature novelty claim.

## 1. Compare physical quantities, with the regulators separate

Use $a$ for the spacetime lattice spacing, $L$ for spatial box size, $m_q$ for a renormalized quark mass, and $\varepsilon = |\pi|$ for the largest step of a Newtonian time partition. Euclidean time extent is taken to infinity to select zero temperature. Write $E_*$ for an energy threshold, $M_* = E_*/c^2$ for a mass, and $\mu_* = E_*/(\hbar c)$ for an inverse correlation length. This avoids the convention in older notes where $m$ sometimes denotes an energy.

For an isotropic reflection-positive lattice theory, on a specified physical channel, normalize the transfer matrix by its vacuum eigenvalue and write

$$T_a = e^{-aH_a/(\hbar c)}, \qquad \delta_a = \frac{aE_*(a)}{\hbar c} = a\mu_*(a).$$

A continuum theory with $0 < E_* < \infty$ therefore has $\delta_a \to 0$. A massless channel also has $\delta_a \to 0$. Their distinction is the limit of $\delta_a/a$, together with convergence of the states and observables that detect it. A strong-coupling lattice gap at fixed bare coupling does not specify that limit.

| Comparison | Quantity to retain in physical units | Required limiting structure |
|---|---|---|
| Pure $SU(3)$ | $0 < E_{\mathrm{YM}} < \infty$ | A nontrivial continuum theory; a gap above its vacuum |

| Comparison | Quantity to retain in physical units | Required limiting structure |
|---|---|---|
| Two-flavour QCD, $m_q = 0$ | $E_\pi = 0$ with nonzero pion spectral weight | Conserved non-singlet axial current and a broken-symmetry vacuum |
| Two-flavour QCD, small $m_q > 0$ | $E_\pi(m_q) > 0$, tending to zero with $m_q$ | Controlled explicit symmetry breaking and the same physical scale |
| Newton comparison | $h_* > 0$ in a specified class of physical records | A record algebra and cost surviving $\varepsilon \to 0$ |

The first three rows use a quantum theory with $\hbar > 0$ already in its definition. The last main goal asks how that positive action structure is justified. The pion supplies a useful check: positive $\hbar$ and a dynamically generated strong-interaction scale can coexist with zero energy gap.

## 2. Pure Yang–Mills: retain coercivity through the limit

The target is a nontrivial four-dimensional $SU(3)$ theory with the Wightman or Osterwalder–Schrader properties, and physical Hamiltonian $H$ satisfying

$$H\Omega = 0, \qquad E_{\mathrm{YM}} := \inf\bigl(\operatorname{spec} H \setminus \{0\}\bigr) \in (0, \infty).$$

One chooses a vacuum sector and proves the corresponding vacuum and clustering properties. This target requires a gap above the vacuum; it does not prescribe every point of the spectrum above the threshold. The official statement explicitly couples existence, axioms and the gap (Jaffe–Witten, §§4–6, passage).

For the lattice route, fix a renormalized physical scale and tune $g_0(a)$ toward zero. A sufficient spectral estimate has the form

$$H_{a,L} - E_{0,a,L} \;\geq\; E_-(1 - P_{0,a,L}), \qquad E_- > 0,$$

uniformly along the trajectory in sufficiently small $a$ and arbitrarily large $L$, on the gauge-invariant Hilbert space. To pass this estimate to a continuum theory one still needs convergence of renormalized local observables, positivity, covariance, regularity and nontriviality. An upper estimate or a surviving finite-energy excitation prevents all non-vacuum states from escaping to infinite energy.

A corresponding correlation route is a bound $C_O(t) \leq K_O e^{-E_- t/\hbar}$ for positive Euclidean time autocorrelations of a dense family of vacuum-subtracted local vectors. The constants must survive the cutoff limit for their chosen renormalizations. One rapidly decaying glueball correlator establishes only a channel statement. Reflection positivity and reconstruction are part of the route from correlations to the physical Hamiltonian; a Langevin relaxation rate by itself is a different object.

The existing H1/H2 route aims to deliver $E_- = (\hbar c/a_*)\min(\gamma_*, \gamma')$. It remains conditional on the blocking and mixing estimates. Construction and nontriviality, called T4 in the obligations map, must remain explicit as well. We choose $L \to \infty$ at fixed $a$ followed by $a \to 0$ along the trajectory as the working order; interchanging these operations requires its own uniform estimates.

**Strong-coupling inputs, corrected 2026-10-02.** The Wilson transfer matrix and the Kogut–Susskind Hamiltonian are distinct operators. For Wilson, the polymer proof gives a volume-uniform rate $4\log[1/(176(e^{6/g^2} - 1))]$ in units $\hbar c/a$ at the sufficient integer bound $g^2 \geq 1059$, on spatial tori with sides at least four. Its exact sufficient KP condition, the strict margin

for a positive rate, and the linearized figure 1056 are distinguished in that proof. The Dobrushin proof gives another Wilson criterion, $g^2 > 24/\log(19/18)$, with $g^2 > 444$ sufficient. Neither sufficient criterion is a phase-transition threshold.

The former explicit KS volume-uniform threshold was withdrawn. Its corrected bound is

$$\Delta_{a,L}^{\mathrm{KS}} \geq \left(\frac{8}{3}g^2 - \frac{12P}{g^2}\right)\frac{\hbar c}{a}, \qquad P = 3(L/a)^3,$$

on the stated periodic cubic lattices, with $L/a \geq 4$. Its volume dependence prevents a uniform numerical threshold. The Yarotsky application and perturbed target region retain existential volume-uniform statements. Their constants and perturbation tolerances do not transfer automatically to Wilson. These fixed-cutoff inputs leave arrival from the weak scaling trajectory and continuum reconstruction open.

## 3. The pion: preserve the symmetry that requires a soft channel

Take $SU(3)$ colour with two degenerate quarks, zero temperature, vacuum angle zero, and no electromagnetism. This fixes what is meant by the comparison with and without a quark mass. Additional flavours or electromagnetism would change the statement.

**Massless case.** Conditional on a local relativistic continuum theory and spontaneous breaking $SU(2)_L \times SU(2)_R \to SU(2)_V$, the Goldstone theorem requires three massless modes with nonzero axial-current matrix elements. These are the pion modes. The theorem's symmetry-breaking premise is a dynamical obligation in QCD; the theorem does not establish it merely from the presence of fermions (Goldstone–Salam–Weinberg 1962, abstract; Leutwyler 1993, §§2–3, passage).

A possible order of limits for the per-flavour order parameter is

$$\Sigma_R = -\lim_{m_q \downarrow 0}\lim_{L\to\infty}\lim_{a\to 0}\langle \bar{u}u\rangle_{R,a,L,m_q} > 0,$$

with a positive mass selecting the vacuum and the physical scale held fixed. The inner continuum limit is taken at fixed physical $L, m_q$. An exactly symmetric finite-volume state has zero symmetry-breaking expectation at zero source; it cannot select the required vacuum by itself. Finite-volume zero modes and the infinite-volume order parameter must be distinguished (Damgaard–Fukaya 2009, introduction, passage). The displayed order is a proposed construction order, with existence at each stage an obligation.

**Nonzero case.** At fixed small positive $m_q$, removing $a$ retains explicit chiral breaking. In the conventional broken-phase low-energy description, put $e_q = m_q c^2$ and let $B_E > 0$ have energy units. Then

$$E_\pi^2 = 2B_E e_q[1 + r(e_q)], \qquad r(e_q) \longrightarrow 0 \quad (e_q \downarrow 0).$$

This is the leading chiral mass relation, conditional on the broken phase and its controlled low-energy expansion; higher orders include chiral logarithms. In particular, $a \to 0$ at fixed $e_q > 0$ is different from $e_q \to 0$ (Leutwyler, *Light Quark Masses*, eq. (1), passage). We use this as a benchmark, without claiming a constructive proof of massive QCD or a lower bound on every channel from this formula.

There is a sharper identity underneath the expansion. Define $\mu_q = m_q c/\hbar$ and $\mu_\pi = E_\pi/(\hbar c)$, both inverse lengths. Assume the renormalized non-singlet Ward identity in compatible Euclidean conventions, away from contact insertions:

$$\partial_\alpha A_\alpha^b = 2\mu_q P^b.$$

With pole amplitudes defined by $\langle 0|A_\alpha^b|\pi^d(k)\rangle = i\delta^{bd}\mathcal{F}_\pi k_\alpha$ and $\langle 0|P^b|\pi^d\rangle = \delta^{bd}\mathcal{G}_\pi$, its one-pion matrix element reads

$$\mu_\pi^2 \mathcal{F}_\pi = 2\mu_q \mathcal{G}_\pi.$$

The conventional continuation fixes the displayed phases. The equation is exact once the current identity and the pion pole exist; deriving the leading mass formula also uses the small-mass behaviour of the residues. A pole with nonzero residue must survive the limit. An unrenormalized divergent pseudoscalar susceptibility is insufficient: contact terms and ultraviolet divergences need separate control.

At finite $a$ the regulator matters. A Ginsparg–Wilson Dirac operator has an exact lattice non-singlet chiral symmetry (Lüscher 1998, abstract). Wilson fermions instead require critical-mass tuning and a restored, renormalized Ward identity (Bali et al. 2019, §2, passage). Exact lattice chirality alone supplies neither spontaneous breaking nor a continuum construction.

For the final paper this comparison earns its place by testing the mechanism. A proposed pure-gauge estimate cannot be carried unchanged to chirally broken QCD if it gaps the pion channel. Colour gauge symmetry and global axial flavour symmetry have different roles. Supersymmetric cancellation of transverse zero-point energies, recorded in G07, is a separate mechanism from QCD's Goldstone pions. Nor does a massless pion imply a massless coloured gluon.

## 4. Newton: retain a record cost while the polygon error vanishes

Use $M$ for the body's mass, constant force $F$, horizontal speed $v$, and comparison duration $T$. The specified Galileo comparison is

$$q_I(t) = (vt, 0), \qquad q_F(t) = (vt, Ft^2/2M),$$
$$A(T) = \frac{vFT^3}{6M}, \qquad \mathcal{A}(T) := \frac{3F}{v}A(T) = \frac{F^2T^3}{2M} = T\Delta E.$$

For a partition $\pi$ of $[0, T]$, the action difference associated with its inscribed chord segments is

$$K_\pi = \frac{F^2}{24M}\sum_j (\Delta t_j)^3, \qquad 0 \le K_\pi \le \frac{F^2T}{24M}\varepsilon^2 \to 0.$$

This follows by $(\Delta t_j)^3 \le \varepsilon^2 \Delta t_j$ and $\sum_j \Delta t_j = T$. In the supplied quantum lift the phase difference is $K_\pi/\hbar$, which also tends to zero at fixed $\hbar > 0$. The polygon note separately gives an ordering phase $\Phi_\pi$ with

$$\hbar\Phi_\pi + 4K_\pi = \frac{\mathcal{A}(T)}{3}.$$

Thus one geometric action defect vanishes while the ordering comparison retains a finite action at fixed $T$. Reading that phase requires a relative-phase experiment. Its derivation already assumes the Weyl relations and positive $\hbar$.

The proposed $h$ gap concerns what a physical record must cost. For a fixed class $\mathfrak{M}_\varepsilon$ of calibrated marks, define

$$\kappa_\varepsilon = \inf_{M_j \in \mathfrak{M}_\varepsilon} \delta_j \Delta_j,$$

where $\delta_j$ is reading error and $\Delta_j$ impulse uncertainty. The class, available preparations, apparatus resources and meaning of the uncertainty must be specified, with feasible informative records of finite cost retained in the limit. A restricted theorem might establish $\inf_{0<\varepsilon<\varepsilon_0} \kappa_\varepsilon \ge \kappa_* > 0$. The general-instrument formulation instead bounds the disturbance functional defined in the disturbance note. These are distinct formulations of record cost; a single-mark product does not by itself establish a universal protocol bound.

The current paper proves such conditional bounds using quantum kinematics or an assumed positive mark trade-off. The main necessity goal is to supply independently justified physical premises that force a positive scale and its universality, without placing that scale in the premise. Identification with $h = 2\pi\hbar$ is a further calibration obligation. Newton's optical fits provide a candidate product $\Lambda p$; the historical argument distinguishes a conditional longitudinal phase count from transverse uncertainty, quantum kinematics and cross-colour universality.

Two refinement operations must stay separate: making more marks within fixed $T$, and asking each shrinking cell to decide the force on its own. The Gaussian result bounds the latter decision window; it permits marks much denser than that window. A positive action cost is compatible with continuous time and with convergence of the polygon. It is also compatible with zero energy gap, as the quantum free particle illustrates.

### 4a. Coherence and the topology of the classical limit

**Gap thesis (user, 2026-10-01).** The proposed necessity claim is that no classical zero branch is continuously reachable. The reachability proof establishes scoped tests of that claim. Its Theorem 1 gives a classical limit for Gaussian preparations and marks. For coherent preparations, Theorem 2's finite-path, nondegenerate stationary-phase setting gives a limit of complete records only when the signed cross amplitudes at every nonzero action difference cancel. Smeared records recover the classical branch sum. The inertial fold and Kepler anomaly examples retain the local crossing and winding hypotheses of Theorems 3–4; they establish no universal persistence at all later times.

An independent coherence premise matters. Theorem 5 proves exactly for quadratic forces that an incoherent mixture of coherent packets, with both position and momentum widths tending to zero, recovers the same classical swarm's branch-weighted coordinate density at every fixed time and regular record value. Weak convergence also holds at critical values. Determinate classical launch data alone therefore do not select the coherent attachment. The fixed coherence-length construction of §3c supplies a nonvanishing crossing-stream contrast when coherence between the distinct launch points is retained as $\hbar \to 0$; that retention is a preparation premise.

The leading fringe visibility is a dimensionless contrast, not an energy threshold or a generated action unit. Complete-record, smearing, preparation and phase-constant limits must be specified separately. Their noncommutation does not identify those limits with the volume, cutoff and coupling limits in §2. A physical law that selects and preserves the required coherence, and calibration of a universal positive action scale, remain obligations.

#### Singular completion is a separate comparison

The Coulomb comparison proves fixed-interval, collision-free semiclassical recovery with a fixed smooth core and separately controlled position/momentum moments. Its relativistic Dirac and Klein–Gordon thresholds depend on operator domain, finite-energy and extension selection. Alternative self-adjoint boundary data can admit every supplied $\hbar > 0$. These are families of chosen point-source theories, not a measured universal action spectrum; critical Klein–Gordon evolution retains an unresolved singular multiplier. The full estimates and primary domain statements belong in that canonical proof rather than a second table here.

## 5. Two elementary criteria for carrying spectral information to a limit

These statements are standard consequences of the spectral theorem and weak convergence of positive measures, included to state exactly what the comparison needs. They are written derivations, with no numerical test and no claim of novelty.

**Proposition 1 (a surviving lower bound).** Suppose renormalized, vacuum-subtracted local vectors $O_n\Omega_n$ have finite positive energy spectral measures $\nu_n$ converging weakly to $\nu$ in a reconstructed limiting theory. If every $\nu_n$ is supported in $[E_-, \infty)$ for one $E_- > 0$, then so is $\nu$.

*Proof.* Every nonnegative continuous test function compactly supported in $[0, E_-)$ integrates to zero against every $\nu_n$, hence against $\nu$. Such test functions exhaust the interval. If the statement holds on a dense family of local vectors orthogonal to the vacuum, the spectral projection of $H$ onto $(0, E_-)$ is zero. A nonzero limiting measure supplies surviving finite-energy spectral content. □

For positive autocorrelations $C_n(t) = \int e^{-Et/\hbar}\, d\nu_n(E)$, a uniform physical decay rate is another way to obtain the support condition. The normalization of the observables and their nonzero limit are part of the hypotheses.

**Proposition 2 (a surviving zero threshold).** Suppose instead that $\nu_n \Rightarrow \nu$, that $\nu(\{0\}) = 0$, and that for every $\eta > 0$ there is $w_\eta > 0$ such that, for all sufficiently large $n$,

$$\nu_n([0, \eta/2]) \geq w_\eta.$$

Then $\nu((0, \eta)) > 0$ for every $\eta > 0$, and the channel's excited energy threshold is zero.

*Proof.* The closed-set inequality for weak convergence gives

$$\nu([0, \eta/2]) \geq \limsup_n \nu_n([0, \eta/2]) \geq w_\eta.$$

Removing the zero atom leaves positive weight in $(0, \eta)$. □

The no-atom hypothesis distinguishes arbitrarily soft excitations from an extra vacuum contribution. It belongs to a selected pure vacuum and a connected observable. The chiral comparison suggests obtaining the required surviving weight from an axial Ward identity and a nonzero order parameter, with the Goldstone theorem supplying the continuum implication.

**Why convergence of eigenvalues alone is too weak.** For $E_0 > 0$ let $\mathrm{d}_x$ denote unit point mass at energy $x$, and set

$$\nu_n = (1 - e^{-n})\mathrm{d}_{E_0} + e^{-n}\mathrm{d}_{E_0/n}.$$

Every $\nu_n$ has lowest energy $E_0/n \to 0$, yet $\nu_n \Rightarrow \mathrm{d}_{E_0}$. The soft state's observable weight disappears. Conversely, a positive gap at each $n$ can collapse with nonvanishing weight. These examples identify the estimates a continuum argument must actually carry. Compact $U(1)$ gauge theory in three dimensions realizes the collapse as a theorem: every lattice has a positive Debye mass, and at fixed coupling the continuum limit is the massless free field, so the uniform $E_-$ of Proposition 1 fails; the series/parallel note, §4 traces the failure to the error density of one refinement step in physical units.

For Newton the analogous surviving object is a calibrated record and its information about the comparison. No spectral measure is supplied by classical mechanics for this purpose. The useful transfer is the proof discipline: compatible observable limits, nonvanishing response, and a bound uniform over admitted refinements. Any stronger equivalence requires a constructed mathematical map.

## 6. The mechanism worth developing across the two main goals

The action-floor note G08 supplies one concrete bridge. A transverse oscillator of frequency $\Omega(x)$ obeys

$$\frac{p_y^2}{2M} + \frac{1}{2}M\Omega(x)^2 y^2 \;\geq\; \frac{1}{2}\hbar\Omega(x).$$

When $\Omega(x)$ grows along a classically escaping valley, this all-state operator bound can confine that valley. Its force comes from the action scale together with the non-abelian commutator

potential. The solved matrix models establish the mechanism in a specified Hamiltonian; extension to four-dimensional gauge theory requires uniform control of the additional modes and renormalization.

The pion comparison asks which flat directions an interaction is allowed to lift. A broken exact global symmetry protects Goldstone directions. Pure Yang–Mills has no corresponding axial flavour symmetry. The absence of that protection makes lifting possible; the positive bound still has to be proved. Finite quark mass then supplies a controlled way to lift the protected direction, which is why both pion cases are useful as orientation.

The candidate unifying question is therefore: **which structures enforce an irreducible cost, which protect a soft direction, and which survive the removal of the regulator?** Quantum kinematics supplies an action unit to both QCD theories. Their different spectra depend on dynamics and symmetry. A successful Newton necessity argument must explain the action structure at the preceding level.

## 7. Consequence for STATE

The retained QCD Ward identity and two spectral-measure criteria explain what must survive a physical limit. A disappearing eigenvalue with vanishing observable weight is insufficient; a supplied action unit does not imply a gapped spectrum. Newton's record problem supplies no spectral measure until one is constructed, so the correspondence is a proof discipline rather than an equivalence theorem. The gauge atlas owns the dimensional/refinement map, and STATE owns the live priorities. Both main goals remain open; this comparison introduces no additional research queue.